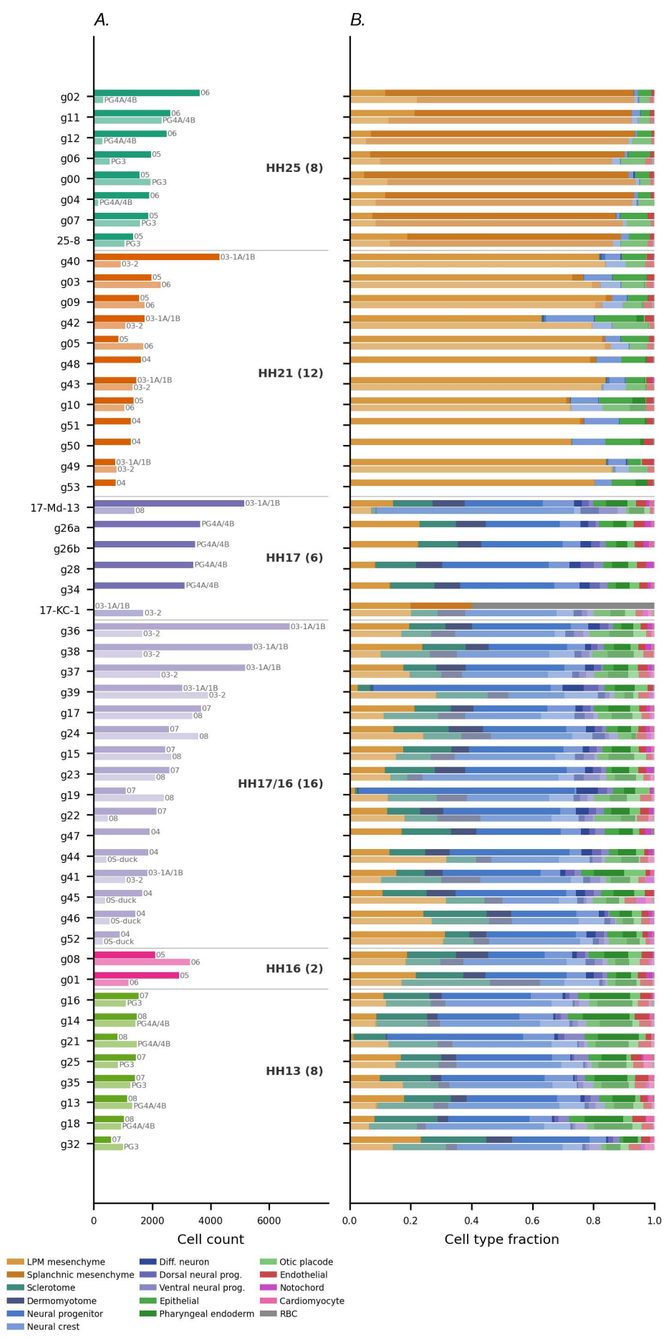
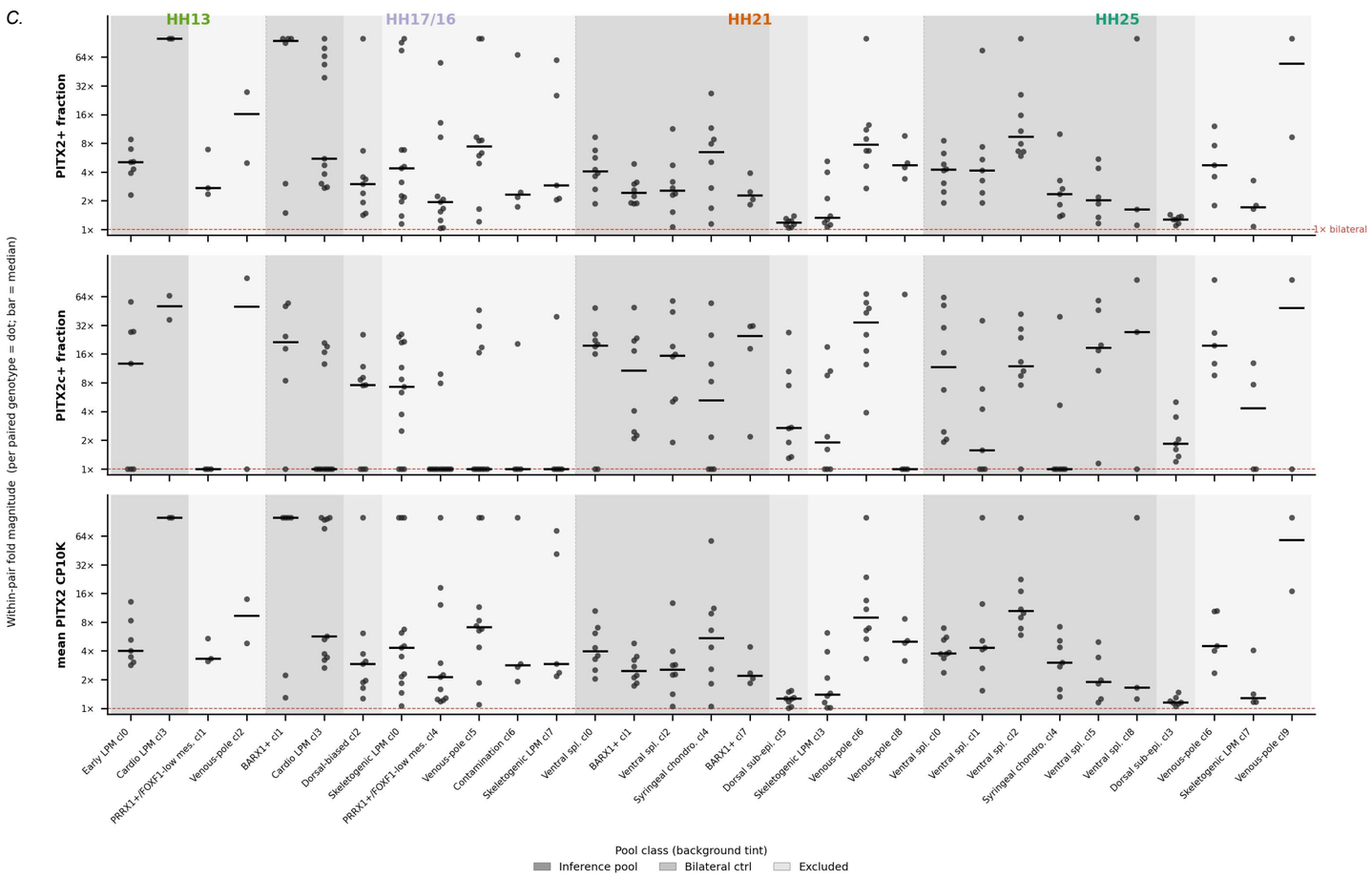
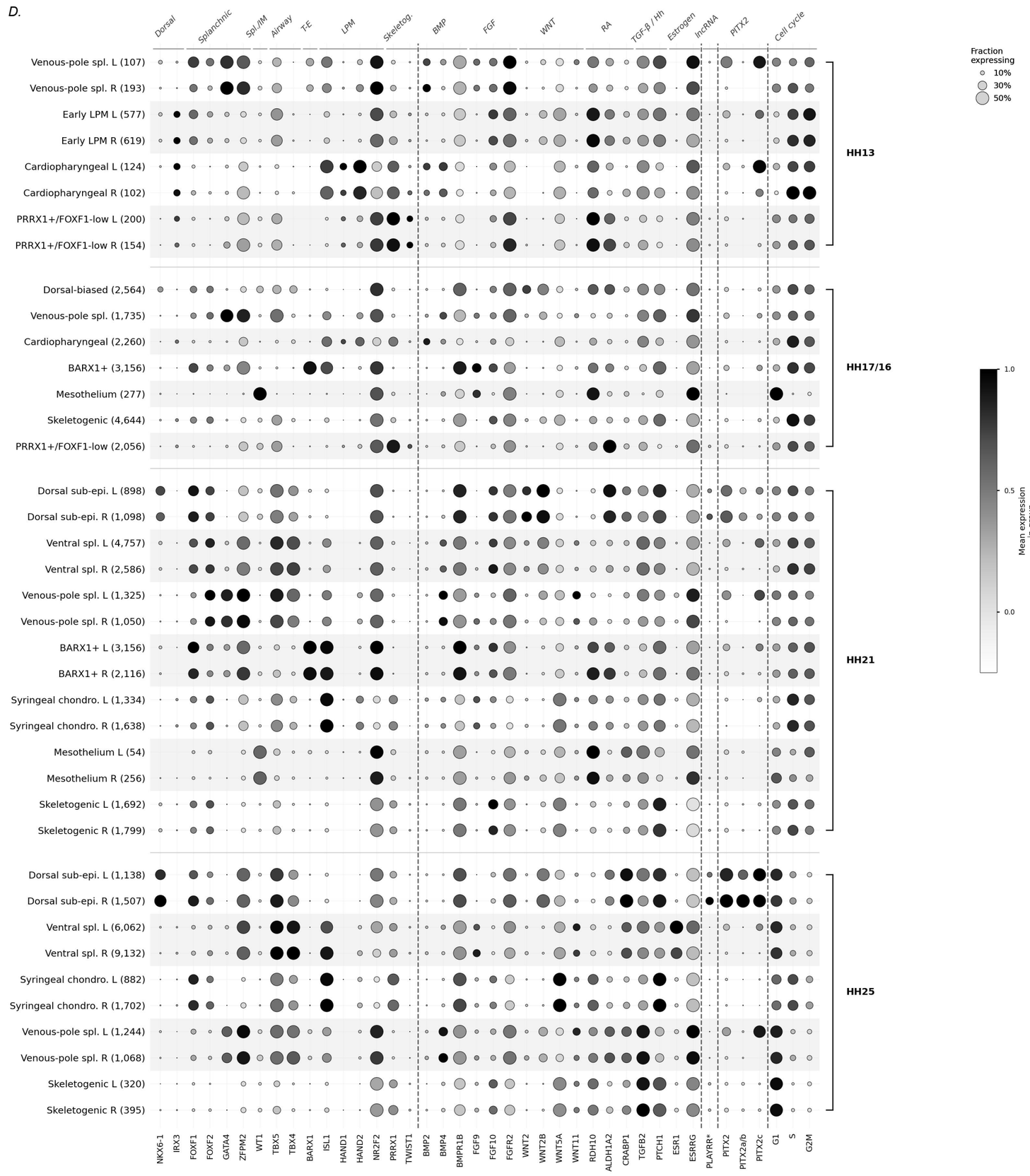
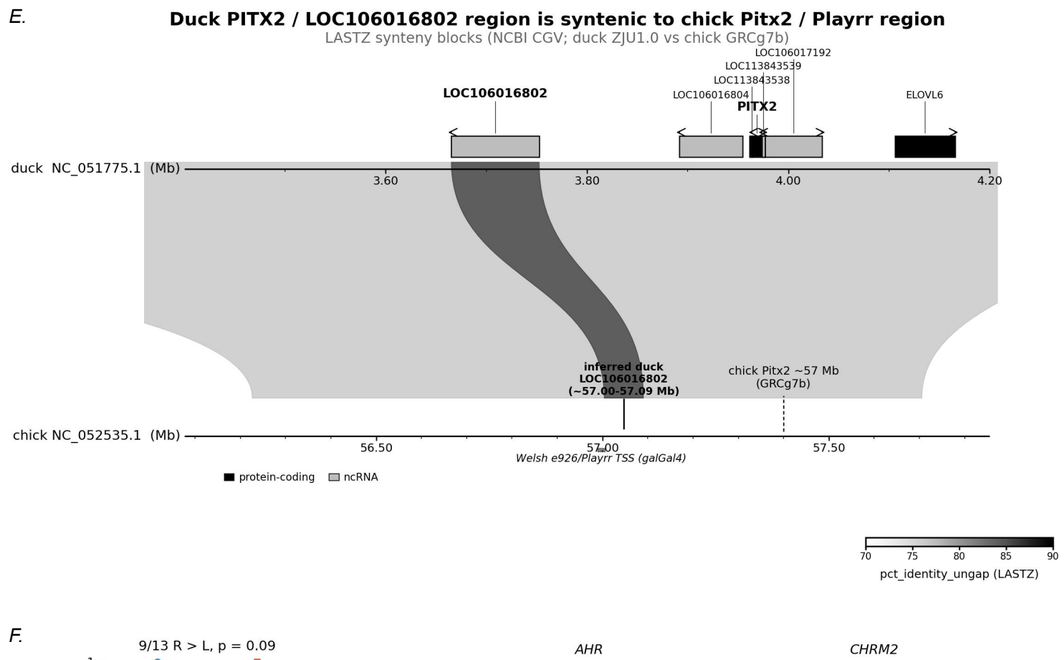

A.
B.
HH25 (8)
HH21 (12)
HH17 (6)
HH17/16 (16)
HH16 (2)
HH13 (8)
Cell count
Cell type fraction
LPM mesenchyme
Splanchnic mesenchyme
Sclerotome
Dermomyotome
Neural progenitor
Neural crest
Diff. neuron
Dorsal neural prog.
Ventral neural prog.
Epithelial
Pharyngeal endoderm
Otic placode
Endothelial
Notochord
Cardiomyocyte
RBC


C.
HH13
HH17/16
HH21
HH25
PITX2+ fraction
PITX2c+ fraction
mean PITX2 CP10K
Within-pair fold magnitude (per paired genotype = dot; bar = median)
1× bilateral
Early LPM cl0
Cardio LPM cl3
PRRX1+/FOXF1-low mes. cl1
Venous-pole cl2
BARX1+ cl1
Cardio LPM cl3
Dorsal-biased cl2
Skeletogenic LPM cl0
PRRX1+/FOXF1-low mes. cl4
Venous-pole cl5
Contamination cl6
Skeletogenic LPM cl7
Ventral spl. cl0
BARX1+ cl1
Ventral spl. cl2
Syringeal chondro. cl4
BARX1+ cl7
Dorsal sub-epi. cl5
Skeletogenic LPM cl3
Venous-pole cl6
Venous-pole cl8
Ventral spl. cl0
Ventral spl. cl1
Ventral spl. cl2
Syringeal chondro. cl4
Ventral spl. cl5
Ventral spl. cl8
Dorsal sub-epi. cl3
Venous-pole cl6
Skeletogenic LPM cl7
Venous-pole cl9
Pool class (background tint):
Inference pool
Bilateral ctrl
Excluded


D.
Dorsal
Splanchnic
Spl./IM
Airway
T-E
LPM
Skeletog.
BMP
FGF
WNT
RA
TGF-β / Hh
Estrogen
lncRNA
PITX2
Cell cycle
HH13
HH17/16
HH21
HH25
Fraction expressing
10%
30%
50%
Mean expression in group
NKX6-1
IRX3
FOXF1
FOXF2
GATA4
ZFPM2
WT1
TBX5
TBX4
BARX1
ISL1
HAND1
HAND2
NR2F2
PRRX1
TWIST1
BMP2
BMP4
BMPR1B
FGF9
FGF10
FGFR2
WNT2
WNT2B
WNT5A
WNT11
RDH10
ALDH1A2
CRABP1
TGFB2
PTCH1
ESR1
ESRRG
PLAYRR*
PITX2
PITX2a/b
PITX2c
G1
S
G2M


E.
Duck PITX2 / LOC106016802 region is syntenic to chick Pitx2 / Playrr region
LASTZ synteny blocks (NCBI CGV; duck ZJU1.0 vs chick GRCg7b)
LOC106016802
LOC106017192
LOC113843539
LOC113843538
LOC106016804
PITX2
ELOVL6
duck NC_051775.1 (Mb)
chick NC_052535.1 (Mb)
inferred duck LOC106016802 (~57.00-57.09 Mb)
chick Pitx2 ~57 Mb (GRCg7b)
Welsh e926/Playrr TSS (galGal4)
protein-coding
ncRNA
pct_identity_ungap (LASTZ)

F.
9/13 R > L, p = 0.09
Cardiopharyngeal LPM share of pool nuclei
Left
Right
AHR
9/13 > 0
11/13 > 0
CHRM2
9/13 > 0
12/13 > 0
Right − Left expression (CPM)
composition
within subcluster